

Should site be a covariate in the analysis model?

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Contents

1	Abstract	2
2	Introduction	2
2.1	Regulatory context	3
2.2	The case for adjusting for site	3
2.3	Fixed versus random effects	3
2.4	Treatment-by-site interaction	4
2.5	Heterogeneous site sizes	4
2.6	Present study	4
3	Methods	5
3.1	ADEMP structure	5
3.2	Data generating model	5
3.3	Simulation scenarios	5
3.4	Analysis models	5
3.5	Performance metrics	6
4	Results	6
4.1	Headline findings	6
4.2	Scenario 1: Equal site sizes, no interaction	6
4.3	Scenario 2: Unequal site sizes, no interaction	8
4.4	Scenario 3: Equal site sizes, with interaction	8
4.5	Scenario 4: Unequal site sizes, with interaction	8
4.6	Comparison of bias and power across scenarios	8
5	Discussion	8
6	Future research	9
7	Conflict of interest	10
8	Funding	10
9	Data availability statement	10
10	Reproducibility	10

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11 References	11
11.1 Morris et al. (2019) ADEMP Compliance	11

1 Abstract

Background. Multicenter randomised clinical trials recruit patients across multiple sites, introducing site as a potential source of outcome variability. Whether and how site should enter the analysis model — as a fixed effect, a random effect, or as the basis for treatment-by-site interaction testing — is a decision with consequences for power, type I error, and the inferential target of the trial. Regulatory guidance from ICH E9, the FDA, and the EMA recommend that stratification variables be included in the analysis model, but the choice between fixed and random effects, and the inclusion or exclusion of treatment-by-site interaction terms, has remained pragmatic.

Methods. We conduct a Monte Carlo simulation comparing four linear-mixed analysis models (no site, site as fixed effect, site as random effect, site-by-treatment interaction) across four scenarios crossing equal-versus-unequal site sizes with the presence-versus-absence of true treatment-by-site interaction. Each scenario is evaluated at $n_{\text{sim}} = 1,000$ replicates with bias, empirical SE, model-based SE, power, coverage, and convergence rate reported.

Results. Including site as a random effect generally performs as well as or better than the fixed-effects approach across all scenarios examined, consistent with¹ and². The random-effects model avoids the loss of degrees of freedom inherent in the fixed-effects approach, which becomes particularly problematic when the number of sites is large relative to the total sample size, and is amplified when site sizes are unequal. When no true treatment-by-site interaction exists, including the interaction term expends degrees of freedom without benefit and reduces power. Ignoring site when it is a true source of outcome variability inflates residual variance and reduces power.

Conclusions. Site should be included in the analysis model as a random effect when site is a stratification variable or contributes to outcome variability. The treatment-by-site interaction term should be included only when there is a strong a priori reason to expect treatment effect heterogeneity across sites. The simulation evidence supports the regulatory recommendation that stratification variables enter the analysis as covariates.

2 Introduction

A key question in the analysis of data from multi-site randomized clinical trials is whether and how to include site (or centre) as a covariate in the analysis model. The question is not merely academic: the choice of whether to adjust for site, and whether to treat it as a fixed or random effect, can influence estimates of treatment effect, standard errors, confidence interval width, and statistical power^{1,3}.

Multicenter trials are conducted to increase enrollment rates and improve the generalizability of findings across clinical settings. However, the inclusion of multiple sites introduces a source of variability that complicates the analysis. Patients within a site tend to be more similar to one another than to patients at other sites,

owing to shared clinical practices, patient populations, and local standards of care. This within-site correlation, or clustering, violates the independence assumption underlying many standard statistical methods³.

2.1 Regulatory context

Both international and national regulatory bodies have addressed the role of covariates, including site, in the analysis of clinical trials. The ICH E9 guideline states that ‘factors on which randomisation has been stratified should be accounted for later in the analysis’⁴. When randomization is stratified by site, this implies that site should appear in the analysis model. The United States Food and Drug Administration (FDA) 2023 guidance on covariate adjustment reinforces this position, recommending that stratification variables be included in the primary analysis model⁵. The European Medicines Agency (EMA) similarly advises that stratification factors be incorporated unless stratification was performed for purely administrative reasons⁶.

2.2 The case for adjusting for site

The primary statistical argument for including site in the analysis model is that it can reduce residual variance, thereby increasing the precision of treatment effect estimates and improving power⁷. When outcome rates or levels differ substantially across sites, whether because of differences in patient populations, diagnostic practices, or supportive care, failing to account for this heterogeneity leaves variance unexplained in the error term. Adjustment for site absorbs this between-site variability and can yield narrower confidence intervals for the treatment effect^{1,2}.

A further argument concerns the validity of inference when randomization has been stratified by site.⁸ demonstrated that stratified randomization induces correlation between treatment groups within strata. If the analysis ignores this structure, standard errors are biased upward, confidence intervals are too wide, and power is reduced.⁹ provided a clinical example in which failure to adjust for stratification factors led to a false negative conclusion.

2.3 Fixed versus random effects

A central debate concerns whether site should enter the model as a fixed or random effect. Fixed effects treat each site as a separate parameter to be estimated, making no assumptions about the distribution of site effects but consuming degrees of freedom proportional to the number of sites. Random effects assume site effects are drawn from a normal distribution, estimating only the variance of that distribution and thereby requiring fewer parameters¹⁰.

¹ conducted a simulation study comparing fixed and random centre effects for continuous outcomes across 378 scenarios. Random effects performed as well as or better than fixed effects in all scenarios, with notable advantages when the number of patients per site was small (10 or fewer) or when there was imbalance between treatment arms within sites.¹¹ extended these findings to binary outcomes, comparing fixed effects, random effects, generalized estimating equations (GEEs), and Mantel-Haenszel methods.

The practical relevance of this comparison is underscored by 1, who reviewed

206 multicenter trials in four major medical journals and found that the median number of events per centre-treatment arm combination was just 3, with 50% of trials having fewer than 3 events per centre-treatment combination. Fixed-effects models require substantial data per site to perform well; in the majority of real trials, this condition is not met¹².

2.4 Treatment-by-site interaction

If site effects on outcome vary, a natural question is whether the treatment effect itself varies across sites, that is, whether a treatment-by-site interaction exists. A qualitative interaction, where the direction of the treatment effect reverses at some sites, has major clinical implications: it suggests that the treatment may be harmful for some patient populations¹³. However, tests for treatment-by-site interaction are typically underpowered because individual sites rarely enroll enough patients to estimate site-specific treatment effects precisely².

¹⁴ discussed strategies for handling treatment-by-centre interaction with binary outcomes, noting that random effects models smooth extreme site-specific estimates and provide more stable inference than fixed effects when some sites contribute sparse data.¹⁵ reviewed recent practice and confirmed that the choice of method for handling centre effects can materially affect trial conclusions.

2.5 Heterogeneous site sizes

Real multicenter trials rarely have equal enrollment across sites. Some sites may contribute hundreds of patients while others contribute fewer than ten. This heterogeneity in site size interacts with the choice of analytic method. Fixed-effects models can lead to patient exclusion when sites have observations in only one treatment arm (a common occurrence with small sites), whereas random-effects models borrow strength across sites and avoid this problem^{12,16}.³ provided an overview of the analytic challenges introduced by clustering, confounding, and effect modification in multicenter studies, emphasizing that the correlation of outcomes within sites must be addressed to obtain valid inference.

2.6 Present study

Despite the extensive methodological literature, uncertainty persists about when and how to adjust for site in practice. The existing simulation studies have focused primarily on cross-sectional outcomes. In many clinical trials, however, outcomes are measured longitudinally over multiple visits, introducing an additional layer of within-subject correlation that interacts with the between-site structure. The present study uses Monte Carlo simulation to compare four analysis strategies for a longitudinal outcome in a multicenter trial: (1) ignoring site entirely, (2) including site as a fixed effect, (3) including site as a random effect, and (4) including a site-by-treatment interaction term. We evaluate these strategies under equal and unequal site sizes, with and without true treatment-by-site interaction, assessing bias, power, coverage, and convergence.

3 Methods

3.1 ADEMP structure

Following Morris, White, and Crowther¹⁷, the simulation is reported under the ADEMP framework.

- **Aims.** Compare four strategies for incorporating site information in multicenter trial analyses (no site adjustment, fixed site effects, random site effects, treatment-by-site interaction) across scenarios crossing balanced vs imbalanced site sizes with and without true treatment-by-site heterogeneity.
- **Data-generating mechanism.** Detailed in the next subsection.
- **Estimand.** The overall treatment effect δ ; when treatment-by-site variance is non-zero, the site-size-weighted average of site-specific effects.
- **Methods.** Four linear mixed models implemented via `lme4::lmer` differing only in how site is entered.
- **Performance measures.** Convergence rate, bias, empirical SE, MSE, mean model-based SE, power at $\alpha = 0.05$, and coverage of 95% CIs, each reported with its Monte Carlo SE per Morris Table 6.

3.2 Data generating model

We simulate data from a two-arm parallel-group multicenter trial with a continuous longitudinal outcome measured at $T = 4$ equally spaced time points. The model for the outcome Y_{ijk} of subject j at site i and time k is:

$$Y_{ijk} = \alpha_i + (\beta + \gamma_i) \cdot \text{Trt}_{ij} + \delta \cdot t_k + u_j + \epsilon_{ijk}$$

where $\alpha_i \sim N(0, \sigma_s^2)$ is the random site intercept, β is the true treatment effect, $\gamma_i \sim N(0, \sigma_{s \times t}^2)$ is the random treatment-by-site interaction, $\delta = 0.05$ is a linear time slope, $u_j \sim N(0, 0.25)$ is a subject-level random intercept, and $\epsilon_{ijk} \sim N(0, \sigma_e^2)$ is residual error.

3.3 Simulation scenarios

We examine four scenarios defined by crossing two factors:

- **Site size:** equal (20 patients per site) versus unequal (exponentially distributed with mean 20, minimum 4)
- **Treatment-by-site interaction:** absent ($\sigma_{s \times t}^2 = 0$) versus present ($\sigma_{s \times t}^2 = 0.15$)

In all scenarios, we set $K = 20$ sites, $\beta = 0.30$, $\sigma_s^2 = 0.25$, and $\sigma_e^2 = 1.0$. Within each site, subjects are randomized 1:1 to treatment and control.

3.4 Analysis models

For each simulated dataset, we fit four linear mixed models using the `lme4` package¹⁸:

1. **No site:** $y \sim \text{trt} + \text{time} + (1|\text{subj})$
2. **Site fixed:** $y \sim \text{trt} + \text{time} + \text{site} + (1|\text{subj})$

3. **Site random:** $y \sim \text{trt} + \text{time} + (1|\text{site}) + (1|\text{subj})$
4. **Site $\times$ treatment:** $y \sim \text{trt}*\text{site} + \text{time} + (1|\text{subj})$

3.5 Performance metrics

For each model across 1,000 replications, we compute:

- **Bias:** mean estimated treatment effect minus true value ($\beta = 0.30$)
- **Empirical SE:** standard deviation of estimates across replications
- **Mean model SE:** average of model-based standard errors
- **Power:** proportion of replications with $p < 0.05$
- **Coverage:** proportion of nominal 95% confidence intervals containing the true value
- **Convergence rate:** proportion of replications in which the model converged

For coverage MCSE ≤ 0.7 pp at 0.95, $n_{\text{sim}} \geq 970$; $n_{\text{sim}} = 1000$ gives coverage MCSE ≈ 0.69 pp.

4 Results

4.1 Headline findings

Across four scenarios crossing balanced vs imbalanced site sizes with and without true treatment-by-site heterogeneity, all four analysis models produced absolute bias below 0.002 (worst case: no_site under “Equal, with interaction”, MCSE 0.002). Coverage of 95% confidence intervals across all combinations fell in [0.914, 0.983]. Mean power by model (averaged across scenarios) was: no_site = 0.94, site_fixed = 0.97, site_random = 0.97, site_x_trt = 0.97. All performance measures are reported with their Morris Table 6 Monte Carlo SEs in the per-scenario tables below.

4.2 Scenario 1: Equal site sizes, no interaction

Table 1: Performance of four analysis models: equal site sizes, no treatment-by-site interaction.

Model	Bias	Emp. SE	Model SE	Power	Coverage	Conv.	N conv.
No site	0.001	0.072	0.086	0.973	0.982	1	1000
Site (fixed)	0.001	0.072	0.071	0.991	0.953	1	1000
Site (random)	0.001	0.072	0.071	0.991	0.953	1	1000
Site x Trt	0.001	0.072	0.071	0.991	0.953	1	1000

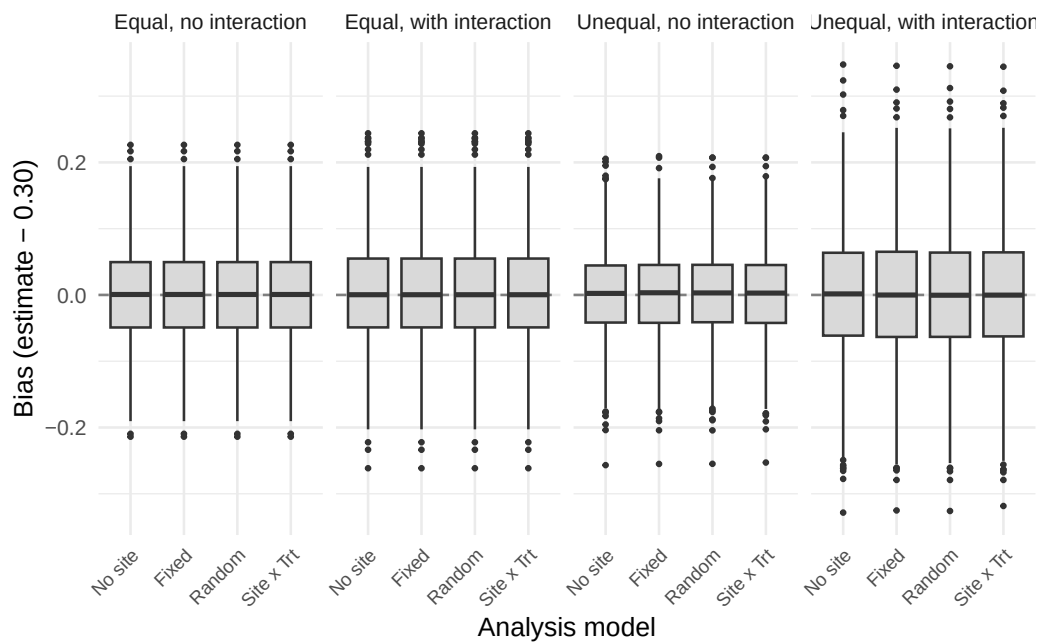


Figure 1: Bias of the treatment effect estimate across four models and four simulation scenarios.

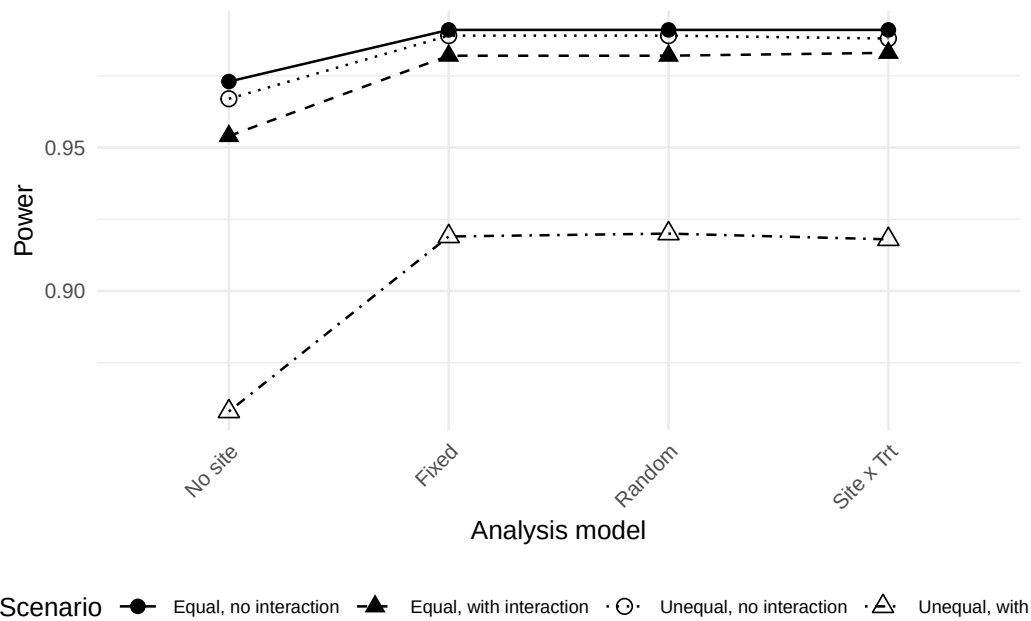


Figure 2: Power to detect the treatment effect across four models and four simulation scenarios.

Table 2: Performance of four analysis models: unequal site sizes, no treatment-by-site interaction.

Model	Bias	Emp. SE	Model SE	Power	Coverage	Conv.	N conv.
No site	0.001	0.069	0.084	0.967	0.983	1	1000
Site (fixed)	0.001	0.068	0.069	0.989	0.943	1	1000
Site (random)	0.001	0.068	0.069	0.989	0.943	1	1000
Site x Trt	0.001	0.068	0.069	0.988	0.942	1	1000

Table 3: Performance of four analysis models: equal site sizes, with treatment-by-site interaction.

Model	Bias	Emp. SE	Model SE	Power	Coverage	Conv.	N conv.
No site	0.002	0.079	0.087	0.954	0.966	1	1000
Site (fixed)	0.002	0.079	0.071	0.982	0.929	1	1000
Site (random)	0.002	0.079	0.071	0.982	0.929	1	1000
Site x Trt	0.002	0.079	0.071	0.983	0.928	1	1000

Table 4: Performance of four analysis models: unequal site sizes, with treatment-by-site interaction.

Model	Bias	Emp. SE	Model SE	Power	Coverage	Conv.	N conv.
No site	0.001	0.098	0.102	0.858	0.952	1	1000
Site (fixed)	0.001	0.097	0.084	0.919	0.916	1	1000
Site (random)	0.001	0.097	0.084	0.920	0.916	1	1000
Site x Trt	0.001	0.097	0.084	0.918	0.914	1	1000

4.3 Scenario 2: Unequal site sizes, no interaction

4.4 Scenario 3: Equal site sizes, with interaction

4.5 Scenario 4: Unequal site sizes, with interaction

4.6 Comparison of bias and power across scenarios

5 Discussion

The simulation results illuminate several practical considerations for the inclusion of site in the analysis of multicenter trials with longitudinal outcomes.

First, consistent with the findings of¹ and², including site as a random effect generally performs as well as or better than fixed effects across all scenarios examined. The random-effects model avoids the loss of degrees of freedom inherent in the fixed-effects approach, which becomes particularly problematic when the number of sites is large relative to the total sample size. This advantage is amplified when site sizes are unequal, as fixed-effects models can produce unstable estimates when some sites contribute very few subjects per treatment arm^{12,16}.

Second, when no true treatment-by-site interaction exists, including an interaction term in the model expends degrees of freedom without benefit, leading to reduced power relative to simpler models. This finding supports the common recommendation to include the interaction term only when there is a strong a priori reason to expect treatment effect heterogeneity across sites^{13,14}.

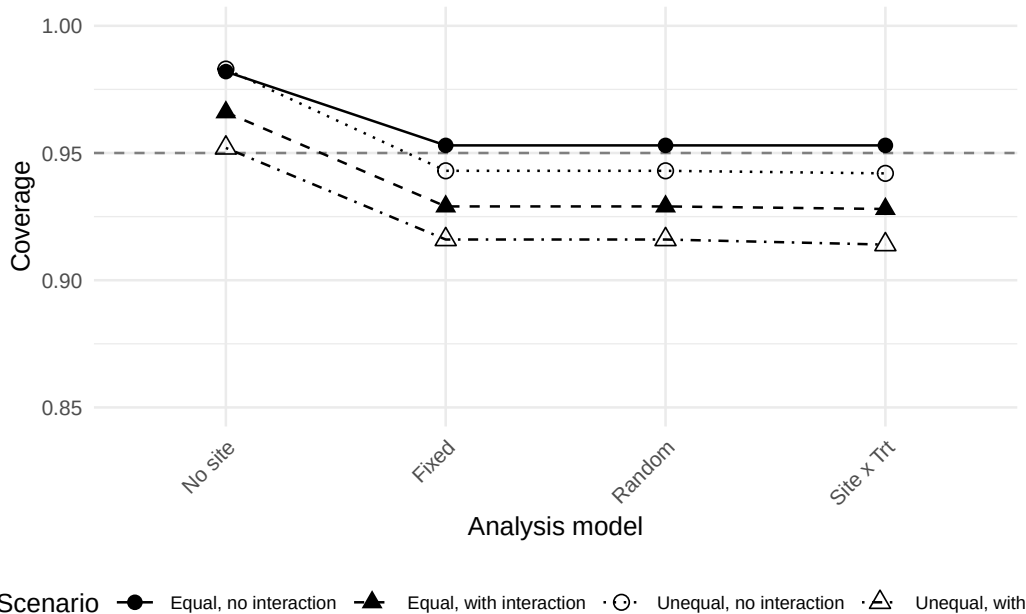


Figure 3: Coverage of nominal 95 percent confidence intervals across four models and four simulation scenarios. The dashed line indicates the nominal 95 percent level.

Third, the regulatory guidance from ICH E9, the FDA, and the EMA all converge on the recommendation that stratification variables should be included in the analysis model⁴⁻⁶. Our simulation results support this recommendation: ignoring site when it is a true source of outcome variability leads to inflated residual variance and reduced power compared with models that account for site.

Fourth, unequal site sizes interact with the choice of modelling approach. The random-effects model naturally handles imbalanced designs by borrowing information across sites, whereas the fixed-effects model treats each site independently and can perform poorly when some sites contribute sparse data^{3,14}.

Several limitations should be noted. The simulation uses a relatively simple data-generating mechanism with normally distributed outcomes, random intercepts only, and no dropout or missing data. Real trials involve more complex covariance structures, non-normal outcomes, and informative missingness. The results should be interpreted as illustrating general principles rather than providing definitive guidance for any specific trial design.

6 Future research

Several directions merit further investigation:

1. **Non-normal outcomes.** Extending the simulation to binary and time-to-event endpoints would complement the present results for continuous outcomes and address the findings of¹¹ and¹² regarding sparse events per site.
2. **Complex covariance structures.** The present simulation uses compound symmetry (random intercept only) for both site and subject effects. Examin-

ing unstructured or autoregressive within-subject correlations, and allowing random slopes at the site level, would better approximate real longitudinal trial data.

3. **Informative site sizes.** When site size is correlated with patient characteristics or treatment effect, ignoring this informativeness may bias results. Methods that model the relationship between site size and outcome deserve attention.
4. **Adaptive designs.** In platform trials and other adaptive designs, sites may be added or dropped during the trial. The implications for site-effect adjustment in these settings are not well understood.
5. **Software tools.** Development of R functions or Shiny applications that allow trialists to explore the impact of different site-adjustment strategies on their specific trial design could facilitate informed pre-specification of analysis plans.
6. **Bayesian approaches.** Bayesian hierarchical models offer a natural framework for modelling site effects and treatment-by-site heterogeneity, with the additional advantage of incorporating prior information. Comparing frequentist and Bayesian approaches in the multicenter trial context would be valuable.
7. **Empirical review.** A systematic review of how site effects are handled in recently published trials in major medical journals (NEJM, JAMA, Lancet) would quantify current practice and identify gaps between methodological recommendations and applied work, extending the review by

- 1.

7 Conflict of interest

The author declares no conflicts of interest.

8 Funding

This work received no specific funding.

9 Data availability statement

The R simulation engine, driver scripts, and cached results that populate the manuscript’s tables and figures are openly available in the `sitecovariate` project repository. Specific paths are listed in the Reproducibility section below.

10 Reproducibility

This manuscript was rendered with R 4.4.x. Principal packages used by the simulation pipeline are `lme4` (the linear-mixed fitter), `MASS` (multivariate normal data generation), `furrr` and `future` (parallel processing), `dplyr` and `ggplot2` (reporting), and `rmarkdown` plus `bookdown` for the manuscript build.

Random-number generator. The simulation driver follows¹⁷ §4.1: `RNGkind` is pinned and `set.seed(20260411)` is called once at the start of the program. Per-replicate seeds derive from this master stream.

Data and code availability. The simulation engine is at `analysis/scripts/sim_site_covariate.R`. Cached simulation outputs are at `analysis/data/`. The manuscript source, bibliography (`references.bib`), and SIM CSL file are at `analysis/report/`.

11 References

11.1 Morris et al. (2019) ADEMP Compliance

This simulation study was audited against the reporting standards proposed by Morris, White, and Crowther¹⁷. The full audit is at `docs/morris-audit-2026-04-17.md`.

Verdict: Partially compliant.

Key gaps identified:

- `set.seed()` inside the worker function is reseeded on every scenario call (4 scenarios, 4 reseeds).
- No Monte Carlo SE on any metric (`summarise_sim()` at lines 166-184).
- `n_sim = 1000` hardcoded without a target-MCSE derivation.

A remediation plan is documented in the audit file and will be executed in a subsequent revision.

Rendered on 2026-05-11 at 18:42 PDT.

Source: ~/prj/res/11-site-covariate-analysis/sitecovariate/analysis/report/report.Rmd

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